ABOUT US

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TRAINOCATE

Application Support and Monitoring

Duration: 3 days

OVERVIEW

Develop L2/L3 application support personnel who can diagnose, monitor, and mitigate production incidents across Linux-hosted applications, MySQL, MS SQL, logs, services, networks, and automation pipelines under time pressure.

OBJECTIVES

- Navigate Linux confidently during production incidents using command-line tools, service management, log inspection, process analysis, disk checks, and network diagnostics.
- Identify slow MySQL queries using slow query logs, process lists, EXPLAIN plans, and runtime evidence.
- Perform MS SQL application-support diagnostics using Query Store, DMVs, wait statistics, blocking analysis, job history, and application correlation.
- Collect evidence before escalation and differentiate application, infrastructure, network, and database root-cause indicators.
- Create Bash scripts that automate evidence collection, log scanning, service checks, database connectivity validation, and maintenance tasks.
- Build a proactive monitoring pipeline that detects application errors, slow queries, deadlocks, disk pressure, service failures, and connectivity problems from logs and command output.
- Communicate incident findings clearly through timelines, findings, probable root cause, business impact, mitigation, and escalation notes.

CONTENT

Module 1: Linux Incident Triage and Application Log Diagnostics

- **Production support mindset and Linux access:** Incident priorities, evidence-first troubleshooting, SSH access, shell prompts, file paths, permissions, sudo, environment variables, and command history.
- **Linux command line for support engineers:** pwd, cd, ls, find, grep, awk, sed, cut, sort, uniq, wc, xargs, less, tail, head, tee, date, tar, and gzip.
- **System services and startup issues:** systemctl, service state, unit files, restart policies, journalctl service logs, and failed Tomcat/application service analysis.
- **Application logs and real-time monitoring:** /var/log, rolling logs, journalctl filters, error patterns, correlation IDs, request IDs, and latency markers.
- **OS resource and disk incidents:** top, ps, free, uptime, df, du, lsof, pidstat, CPU, memory, disk utilization, log growth, and file handles.
- **Network and database connectivity from Linux:** ss, curl, nc, ping, traceroute, nslookup, dig, firewall indicators, DNS checks, TLS/certificate symptoms, and SELinux indicators.
- **Hands-on activities:** Troubleshoot a Tomcat startup failure, investigate application log errors, resolve a disk utilization issue, validate database connectivity, and package incident evidence for escalation.

Lab Output

- Linux incident checklist covering service, process, logs, disk, memory, CPU, and connectivity.
- One evidence bundle containing logs, command output, timestamps, and a short incident summary.
- A basic Bash script that collects system status, service status, recent errors, and disk utilization.

Module 2: Database Performance Diagnostics for MySQL and MS SQL

- **MySQL support diagnostics on Linux:** mysql client basics, service validation, SHOW PROCESSLIST, connection counts, variables, error logs, slow query log configuration, and safe read-only checks.
- **Identifying and analyzing slow MySQL queries:** slow_query_log, long_query_time, mysqldumpslow, rows examined, lock time, query frequency, EXPLAIN, missing indexes, and unsafe assumptions.
- **MySQL contention and reliability incidents:** Deadlocks, InnoDB status, blocked sessions, replication delay, read/write routing, relay logs, and reporting freshness.
- **MS SQL application support diagnostics:** DMVs, active requests, session waits, blocking chains, expensive queries, execution plans, Query Store, and SQL Agent job failures.

- **Blocking, wait stats, and execution plan regression:** sys.dm_os_wait_stats, sys.dm_exec_requests, blocking_session_id, head blocker, statistics, plan regression, and deployment correlation.
- **Cross-correlation between application and database:** Request IDs, user impact, app latency, SQL duration, connection pool exhaustion, retries, and API/downstream dependency checks.
- **Hands-on activities:** Investigate a MySQL-backed application outage, summarize slow queries, analyze a deadlock, diagnose SQL Server high CPU, find a blocking root cause, and build an application/database incident timeline.

Lab Output

- MySQL slow-query triage report with top slow queries, probable cause, evidence, and next action.
- MS SQL incident worksheet covering waits, blocking, Query Store findings, job failure status, and application correlation.
- A database health-check script outline that validates connectivity, captures safe read-only diagnostics, and stores timestamped output.

Module 3: Automation, Proactive Monitoring Pipelines, and Major Incident Simulation

- **Bash scripting for application support:** Variables, command substitution, functions, exit codes, conditionals, loops, arguments, logging, cron, and safe permissions.
- **Automated evidence collection:** Timestamped folders, command output capture, log slicing by time window, masking sensitive data, packaging evidence, and runbook-friendly output.
- **Log parsing and slow query detection pipelines:** grep, awk, sed, CSV output, thresholds, error-rate counts, latency pattern detection, and slow query summaries.
- **Proactive health checks and maintenance scripts:** Service health, endpoint checks, disk thresholds, log cleanup rules, backup/log retention, replication checks, SQL Agent/job status checks, and safe automation boundaries.
- **Integrated P1/P2 incident simulation:** Casino loyalty outage, online gaming delayed transactions, payment integration failure, Linux CPU spike, database slowness, and major incident communication.
- **Capstone review and operational handover:** RCA structure, evidence quality, escalation notes, monitoring backlog, technical recommendation, and training assessment.
- **Hands-on activities:** Create an evidence collector, build a slow-query and log-monitoring pipeline, automate deployment health checks, run a war-room simulation, and present findings with recommended improvements.

Lab Output

- Bash evidence collection script for application, Linux, and database symptoms.
- Proactive log and slow-query monitoring pipeline with threshold-based output.
- Deployment/application health-check script suitable for cron or runbook execution.
- Final incident report containing timeline, business impact, evidence, probable root cause, mitigation, and escalation recommendation.

HANDS-ON SCENARIOS COVERED

- **Application layer + MS SQL:** IIS service suddenly slow, high SQL CPU, blocking chain, execution plan/statistics regression, SQL Agent job failure.
- **Application layer + MySQL on Red Hat Linux:** application down while MySQL service is running, slow player transactions, intermittent login failures, deadlocks during peak hours, replication delay affecting reports.
- **Red Hat Linux:** Tomcat service not starting, disk utilization causing application failure, application cannot connect to database, high CPU application process, DNS/firewall connectivity issues.
- **Integrated simulations:** casino loyalty outage, online gaming transaction delay, third-party payment integration failure, major incident war room, proactive evidence collection, and application health checks.

TOOLS AND COMMANDS PRACTICED

- **Linux:** ssh, sudo, systemctl, journalctl, tail, less, grep, awk, sed, find, tar, gzip, top, ps, free, uptime, df, du, lsof, ss, curl, nc, nslookup, dig, cron.
- **MySQL:** mysql client, SHOW PROCESSLIST, SHOW VARIABLES, slow_query_log, long_query_time, mysqldumpslow, EXPLAIN, SHOW ENGINE INNODB STATUS, replication status checks.
- **MS SQL:** Query Store, SQL Agent job history, sys.dm_os_wait_stats, sys.dm_exec_requests, sys.dm_exec_sessions, sys.dm_exec_query_stats, execution plans, blocking-session analysis.
- **Automation:** Bash functions, scheduled scripts, timestamped evidence folders, threshold checks, log parsing pipelines, health summaries, and runbook outputs.